

04 Rolling Validation Selection Error Rate (Fixed Window)

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Goal (fixed window)

One-step-ahead **fixed** rolling validation with training window $w = 50$: for $t = 1, \dots, n - w$, fit on rows $t:(t + w - 1)$ and predict row $t + w$. Mean squared prediction error is averaged over $n - w$ steps. We compare a **linear** model (M1) and **degree-3 polynomial** (M3), select the model with lower mean RV MSE (tie $\rightarrow$ linear), and report the Monte Carlo **selection error rate** vs n on 100, 150, ..., 1000 (step 50), with 1,000 replications per n .

Linear truth

Simulation settings

Table 1: Simulation settings (fixed window $w = 50$, linear truth).

setting	value
seed	2026
Monte Carlo replications per n	500
Sample sizes (grid)	100, 150, ..., 1000 (step 50)
truth type	linear
beta0	1
beta1	2
beta2	NA
beta3	NA
noise type	gaussian
sigma	4

Error rate summary

Table 2: Monte Carlo mean selection-error rate (truth is linear; error = poly chosen) and approximate MC standard error.

n	error_rate	mc_se
100	0.286	0.02021
200	0.146	0.01579
300	0.086	0.01254
400	0.060	0.01062
500	0.042	0.00897
600	0.018	0.00595
700	0.024	0.00684
800	0.018	0.00595
900	0.004	0.00282
1000	0.002	0.00200

Error rate vs sample size

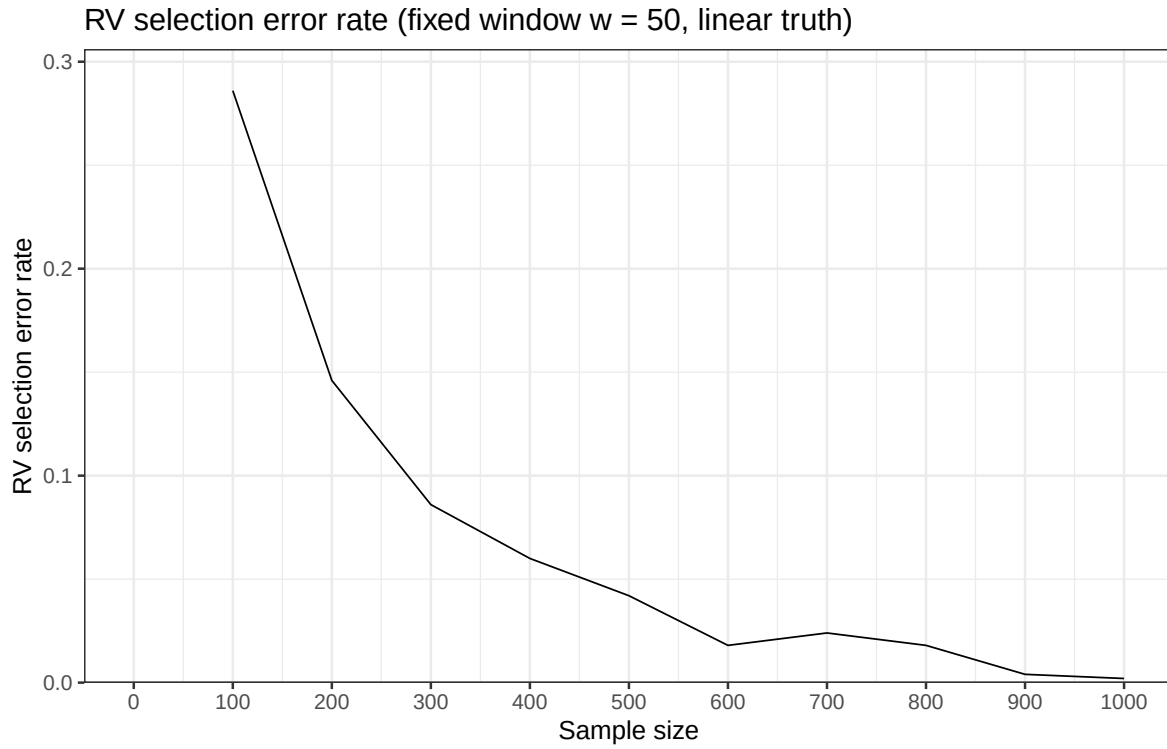


Figure 1: RV selection error rate under linear truth (fixed window $w = 50$).

Polynomial truth

Simulation settings

Table 3: Simulation settings (fixed window $w = 50$, polynomial truth).

setting	value
seed	2026
Monte Carlo replications per n	500
Sample sizes (grid)	100, 150, ..., 1000 (step 50)
truth type	polynomial
beta0	1
beta1	2
beta2	1

setting	value
beta3	1
noise type	gaussian
sigma	4

Error rate summary

Table 4: Monte Carlo mean selection-error rate (truth is polynomial; error = linear chosen) and approximate MC standard error.

n	error_rate	mc_se
100	0.650	0.02133
200	0.822	0.01711
300	0.898	0.01353
400	0.918	0.01227
500	0.958	0.00897
600	0.976	0.00684
700	0.976	0.00684
800	0.982	0.00595
900	0.996	0.00282
1000	0.996	0.00282

Error rate vs sample size

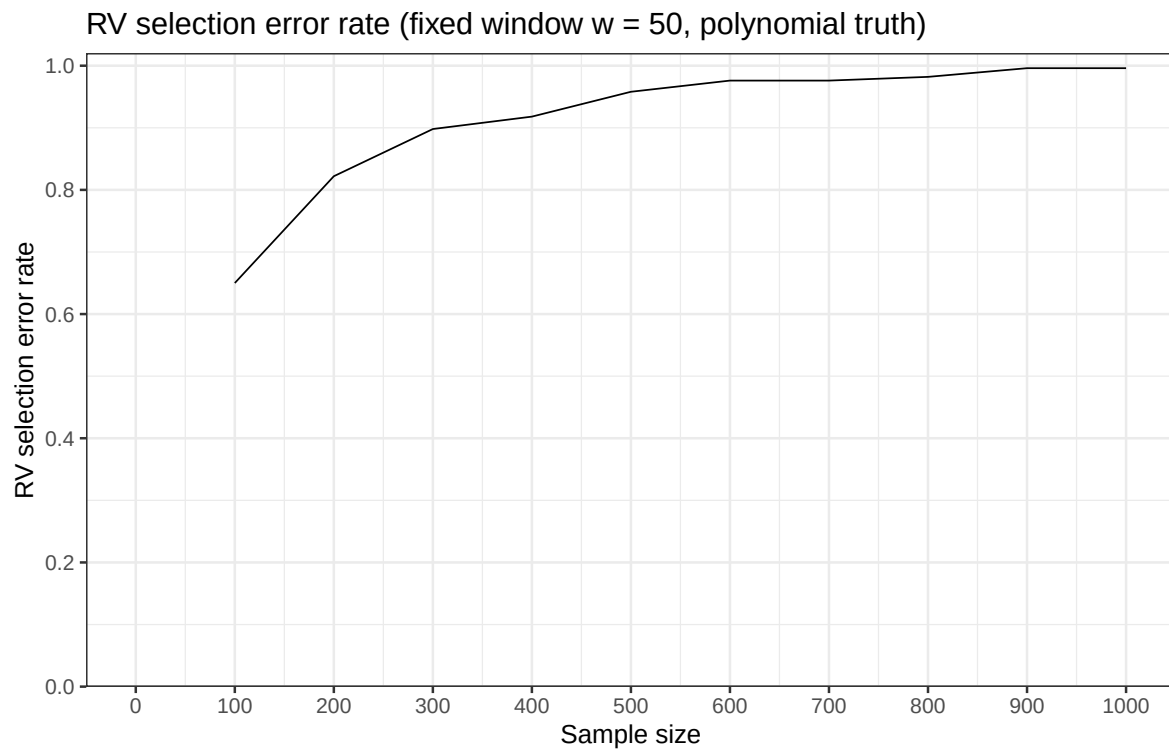


Figure 2: RV selection error rate under polynomial truth (fixed window $w = 50$).